



(SQL) Subqueries

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Objectives

- After completing this lesson, you should be able to do the following:
 - Describe the types of problems that subqueries can solve
 - Define subqueries
 - List the types of subqueries
 - Write single-row and multiple-row subqueries

Using a Subquery to Solve a Problem

- “Who has a salary greater than Jones’?”

Main Query



“Which employees have a salary greater than Jones’ salary?”

Subquery



“What is Jones’ salary?”



Subqueries

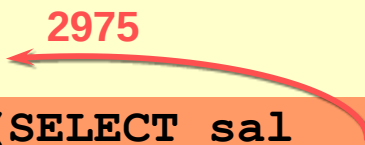
- Query within a query is called **subquery**.

```
SELECT    select_list
FROM      table
WHERE     expr operator
          (SELECT    select_list
           FROM      table);
```

- The subquery (inner query) executes once before the main query.
- The result of the subquery is used by the main query (outer query).
- Bottom-up or inward-outward parsing
- The subquery can also be placed in the **HAVING**, **FROM** clause of the outer query.

Using a Subquery

```
SQL> SELECT  ename
      2  FROM    emp
      3  WHERE   sal >
      4          (SELECT  sal
      5                  FROM    emp
      6                  WHERE   empno=7566) ;
```



ENAME

KING

FORD

SCOTT

Exercise

- List all employees with salary greater than the average salary of the employees.
- List all employees with salary greater than the average salary of the employees of a certain job/department.
- Find the name of the highest paid employee.
- Find the name of the most senior employee.
- List all employees with the same job/department as that of a given employee.

Exercise

- List all employees with salary greater than the average salary of the employees.
- List all employees with salary greater than the average salary of the employees of a certain job/department.
- Find the name of the highest paid employee.
- Find the name of the most senior employee.
- List all employees with the same job/department as that of a given employee.

Types of Subqueries

1. Single-row subquery

- Returns zero or one row to the outer SQL statement. A special case called scalar subquery contains exactly one column.

2. Multiple-row subquery

- Returns one or more rows to the outer SQL statement.
- In addition, there are three subtypes of subqueries that may return single or multiple rows:

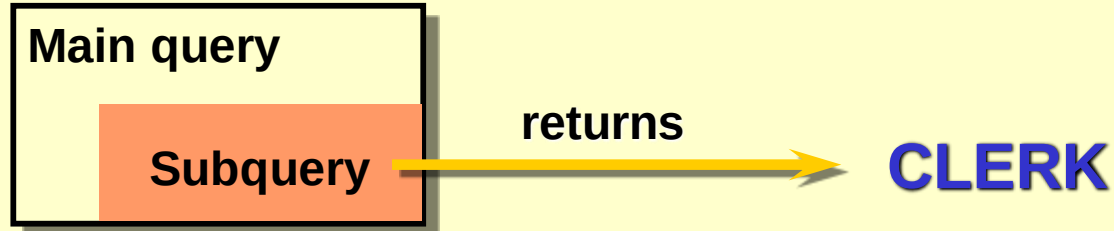
a) **Multiple-column subqueries** return more than one column to the outer SQL statement.

b) **Correlated subqueries** reference one or more columns in the outer SQL statement.

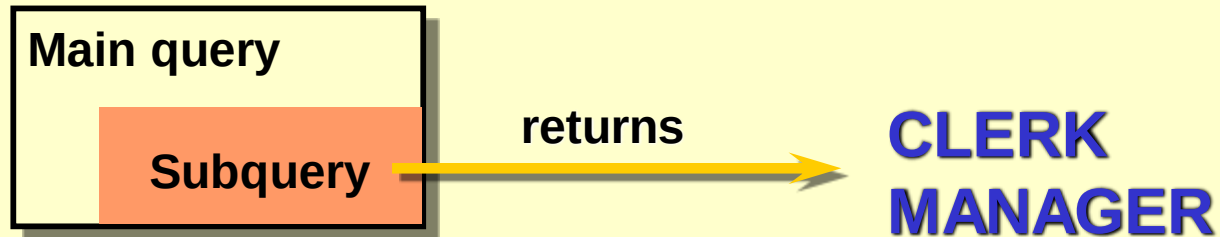
c) **Nested subqueries** are placed within another subquery.

Types of Subqueries

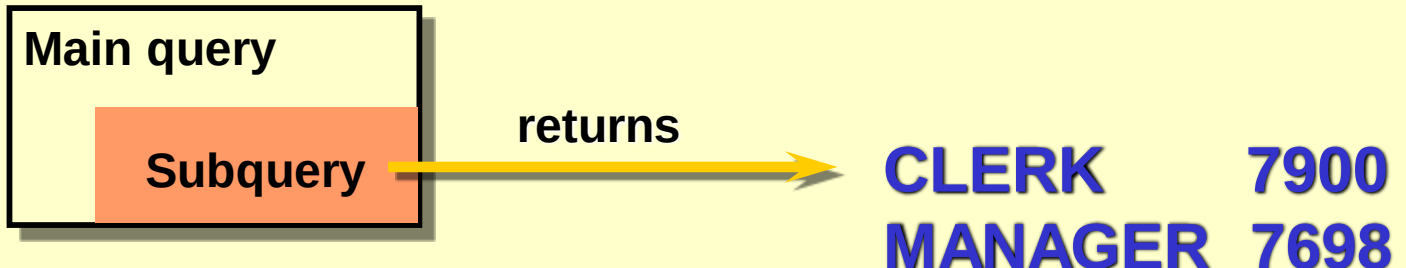
- Single-row subquery



- Multiple-row subquery



- Multiple-column subquery



Guidelines for Using Subqueries

- Enclose subqueries in parentheses.
- Place subqueries on the right side of the comparison operator.
- ORDER BY clause is mostly not required in a subquery.
- Use single-row operators with single-row subqueries.
- Use multiple-row operators (IN,ANY,ALL) with multiple-row subqueries.

Single-Row Subqueries

- Return only one row
- Use single-row comparison operators

Operator	Meaning
=	Equal to
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
<>	Not equal to

Executing Single-Row Subqueries

```
SQL> SELECT      ename, job
  2  FROM        emp
  3  WHERE       job =
  4              (SELECT      job
  5              FROM        emp
  6              WHERE       empno = 7369)
  7  AND        sal >
  8              (SELECT      sal
  9              FROM        emp
 10              WHERE       empno = 7876) ;
```

CLERK

1100

ENAME	JOB
-----	-----
MILLER	CLERK

Using Group Functions in a Subquery (Scalar Subquery)

- Find the lowest paid employee

```
SQL> SELECT  ename, job, sal
2  FROM      emp
3  WHERE      sal =
4              (SELECT  MIN(sal)
5              FROM      emp) ;
```



800

ENAME	JOB	SAL
-----	-----	-----
SMITH	CLERK	800

Finding second highest sal

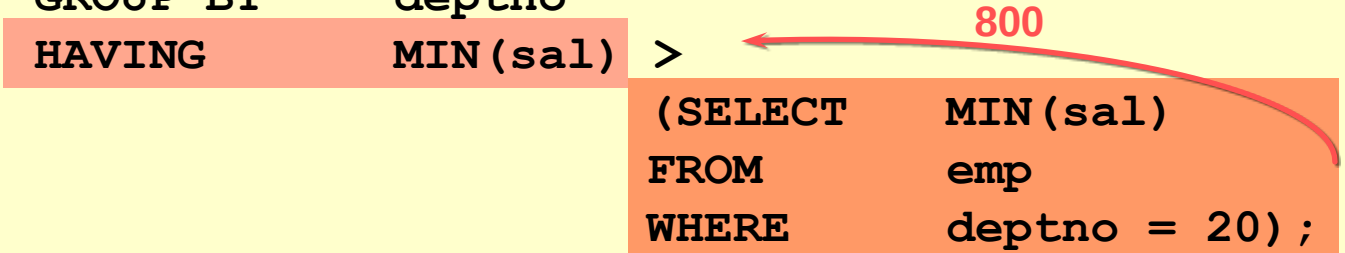
```
SQL> SELECT MAX(sal) FROM emp
2  WHERE sal < (SELECT MAX(sal) FROM emp);
```

```
SQL> SELECT  ename, job, sal
2  FROM      emp
3  WHERE     sal =
4             (SELECT MAX(sal) FROM emp
5  WHERE sal < (SELECT MAX(sal) FROM emp));
```

HAVING Clause with Subqueries

- The Oracle Server executes subqueries first.
- The Oracle Server returns results into the HAVING clause of the main query.

```
SQL> SELECT      deptno, MIN(sal)
2  FROM          emp
3  GROUP BY      deptno
4  HAVING         MIN(sal) >
5                (SELECT      MIN(sal)
6                  FROM        emp
7                  WHERE        deptno = 20) ;
```



- Find the name of the department with the highest/lowest number of employees.

HAVING Clause with Subqueries

- List job, average salary of those jobs whose average salary is greater than the minimum average of different jobs.

```
SELECT JOB, AVG(SAL) FROM EMP  
GROUP BY JOB  
HAVING AVG(SAL) >  
(SELECT MIN(AVG(SAL)) FROM EMP  
GROUP BY JOB)
```


What Is Wrong with This Statement?

```
SQL> SELECT empno, ename
```

```
2 FROM emp
```

```
3 WHERE sal =
```

```
4
```

```
5
```

```
6
```

```
(SELECT MIN(sal)
```

```
FROM emp
```

```
GROUP BY deptno);
```

Single-row operator with
multiple-row subquery

```
ERROR:
```

```
ORA-01427: single-row subquery returns more than  
one row
```

```
no rows selected
```

Will This Statement Work?

```
SQL> SELECT  ename, job
      2  FROM    emp
      3  WHERE   job =
      4          (SELECT job
      5              FROM    emp
      6              WHERE   ename= ' SMYTHE ' ) ;
```

no rows selected

Subquery returns no values

Multiple-Row Subqueries

- Return more than one row
- Use multiple-row comparison operators

Operator	Meaning
IN	Equal to any member in the list
ANY	Compare value to each value returned by the subquery. The overall result is TRUE if it is TRUE for ANY of the values returned by the subquery.
ALL	Compare value to every value returned by the subquery. The overall result is TRUE only if it is TRUE for ALL of the values returned by the subquery.

- ANY, ALL must always be preceded by any of the relational operators (<, <=, >, >=, =, <>)

Using IN Operator in Multiple-Row Subqueries

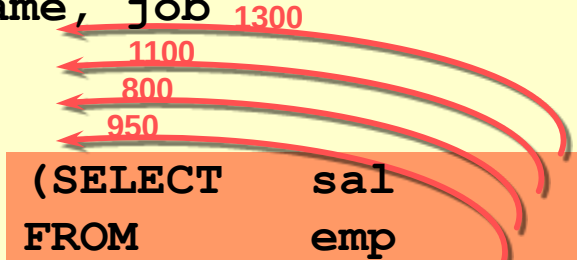
```
SQL> SELECT  empno, ename, job
  2  FROM      emp
  3  WHERE     sal IN      (SELECT  MIN(sal)
  4                               FROM      emp
  5                               GROUP BY deptno);
```

```
SQL> SELECT  empno, sal, deptno
  2  FROM      emp
  3  WHERE     sal < IN (800, 950, 1300);
```

EMPNO	ENAME	JOB
7369	SMITH	CLERK
7900	JAMES	CLERK
7934	MILLER	CLERK

Using ANY Operator in Multiple-Row Subqueries

```
SQL> SELECT empno, ename, job 1300
      2 FROM emp 1100
      3 WHERE sal < ANY 800
      4 (SELECT sal 950
      5 FROM emp
      6 WHERE job = 'CLERK')
      7 AND job <> 'CLERK';
```



EMPNO	ENAME	JOB
7654	MARTIN	SALESMAN
7521	WARD	SALESMAN

Using ALL Operator in Multiple-Row Subqueries

```
SQL> SELECT empno, ename, job
2 FROM emp
3 WHERE sal > ALL
4 (SELECT avg(sal)
5 FROM emp
6 GROUP BY deptno);
```

1566.6667
2175
2916.6667

EMPNO	ENAME	JOB
7839	KING	PRESIDENT
7566	JONES	MANAGER
7902	FORD	ANALYST
7788	SCOTT	ANALYST

Null Values in a Subquery

- List all employees who do not have any subordinates.

```
SQL> SELECT    e.ename
      2  FROM      emp e
      3  WHERE     e.empno NOT IN
      4              (SELECT manager.mgr
      5                  FROM      emp manager);
no rows selected.
```

- One of the values returned by the inner query is **NULL**, and hence now rows are returned.
- The comparison with a **NULL** results in **NULL**.
- **NOT IN** operator is equivalent to **<>ALL**.

Null Values in a Subquery

- List all employees who do not have any subordinates.

```
SELECT    e.ename,e.job FROM emp e
WHERE     e.empno NOT IN
          (SELECT m.mgr FROM emp m
           WHERE m.mgr IS NOT NULL);
```

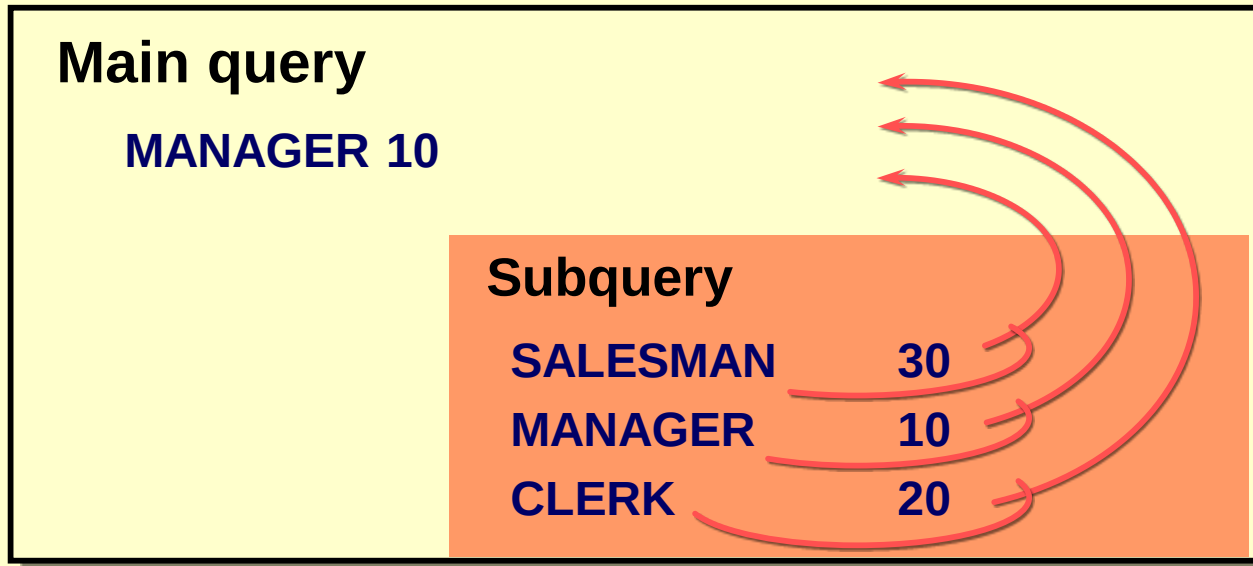
```
SELECT    e.ename,e.job FROM emp e
WHERE     NOT EXISTS
          (SELECT empno FROM emp m
           WHERE m.mgr = e.empno);
```

ENAME	JOB
SMITH	CLERK
ALLEN	SALESMAN
WARD	SALESMAN
MARTIN	SALESMAN
TURNER	SALESMAN
ADAMS	CLERK
JAMES	CLERK
MILLER	CLERK

Multiple-Column Subqueries

- After completing this lesson, you should be able to do the following:
 - Write a multiple-column subquery
 - Describe and explain the behavior of subqueries when null values are retrieved
 - Write a subquery in a FROM clause

Multiple-Column Subqueries



Main query compares to Values from a multiple-row and multiple-column subquery

MANAGER 10

SALESMAN 30
MANAGER 10
CLERK 20

Using Multiple-Column Subqueries

- Display the order number, product number, and quantity of any item in which the product number and quantity match **both** the product number and quantity of an item in order 605.

```
SQL> SELECT  ordid, prodid, qty
  2  FROM      item
  3  WHERE     (prodid, qty) IN
  4              (SELECT prodid, qty
  5                  FROM      item
  6                  WHERE     ordid = 605)
  7  AND        ordid <> 605;
```

Column Comparisons

Pairwise

PRODID		QTY
101863		100
100861	↔	100
102130	↔	10
100890		5
100870	↔	500
101860		50

Nonpairwise

PRODID		QTY
101863	↗	100
100861	↔ ↘	100
102130	↔ ↘	10
100890	↔ ↘	5
100870	↔ ↘	500
101860		50

Nonpairwise Comparison Subquery

- Display the order number, product number, and quantity of any item in which the product number and quantity match any product number and any quantity of an item in order 605.

```
SQL> SELECT  ordid, prodid, qty
2  FROM      item
3  WHERE     prodid IN  (SELECT      prodid
4                          FROM        item
5                          WHERE       ordid = 605)
6  AND       qty      IN  (SELECT      qty
7                          FROM        item
8                          WHERE       ordid = 605)
9  AND       ordid <> 605;
```

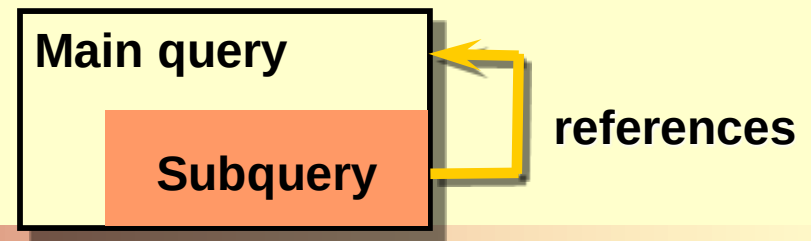
Nonpairwise Subquery

ORDID	PRODID	QTY
609	100870	5
616	100861	10
616	102130	10
621	100861	10
618	100870	10
618	100861	50
616	100870	50
617	100861	100
619	102130	100
615	100870	100
617	101860	100
621	100870	100
617	102130	100

. . .

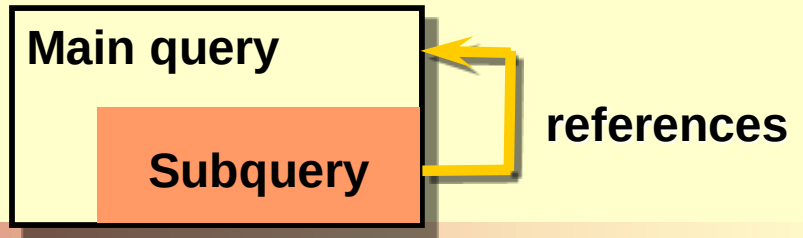
16 rows selected.

Correlated Subquery



- A correlated subquery references one or more columns in the outer SQL statement.
- The inner query uses a table alias of outer query.
- Unlike a simple subquery that is executed only once, a correlated subquery is executed for more than once.
- The inner query is evaluated against each row of the outer query.
- Top-down or outward-inward parsing
- `select * from dept where not exists`
(`select * from emp where emp.deptno = dept.deptno`)

Correlated Subquery

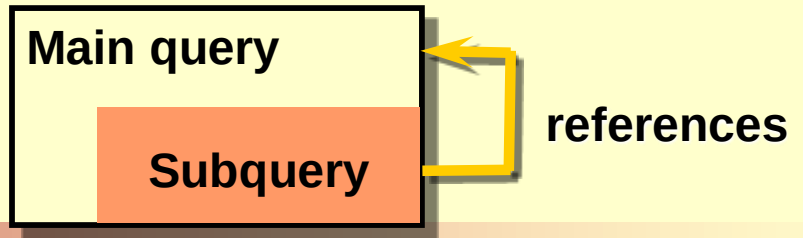


- Subquery in SELECT clause

```
SELECT ename, job, sal,  
       (SELECT ROUND(AVG(sal)) from emp e2  
        WHERE e1.job=e2.job) "avg-job-sal"  
FROM emp e1
```

ENAME	JOB	SAL	avg-job-sal
KING	PRESIDENT	5000	5000
BLAKE	MANAGER	2850	2758
CLARK	MANAGER	2450	2758
JONES	MANAGER	2975	2758
SCOTT	ANALYST	3000	3000
FORD	ANALYST	3000	3000
SMITH	CLERK	800	1038

Correlated Subquery



- Subquery in FROM clause (**inline views**)

```
SELECT  e1.ename, e1.job, e1.sal, e2.salavg
FROM    emp e1,
SELECT  job, round(avg(sal)) salavg
FROM    emp GROUP BY job) e2
WHERE   e1.job = e2.job
```

ENAME	JOB	SAL	avg-job-sal
KING	PRESIDENT	5000	5000
BLAKE	MANAGER	2850	2758
CLARK	MANAGER	2450	2758
JONES	MANAGER	2975	2758
SCOTT	ANALYST	3000	3000
FORD	ANALYST	3000	3000
SMITH	CLERK	800	1038

Using a Subquery in the FROM Clause

```
SQL> SELECT  a.ename, a.sal, a.deptno, b.salavg
  2  FROM    emp a, (SELECT  deptno, avg(sal) salavg
  3                      FROM    emp
  4                      GROUP BY deptno) b
  5  WHERE    a.deptno = b.deptno
  6  AND      a.sal > b.salavg;
```

ENAME	SAL	DEPTNO	SALAVG
KING	5000	10	2916.6667
JONES	2975	20	2175
SCOTT	3000	20	2175
...			

6 rows selected.

Correlated Subquery

- Finding a department without an employee

```
SELECT * from dept
WHERE NOT EXISTS
(SELECT * from emp where emp.deptno=dept.deptno)
```

- Finding an employee with Nth highest salary

```
SELECT * FROM Emp Emp1
WHERE (N-1) = (
SELECT COUNT(DISTINCT Emp2.Sal)
FROM Emp Emp2
WHERE Emp2.Sal > Emp1.Sal);
```

Thank you for your attention.

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